

A Machine Learning Approach to Orbital Object Classification Using Random Forest Algorithms

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Abstract

This paper documents the implementation of a Random Forest machine learning model designed to classify orbiting bodies as either space debris or active satellites. Utilizing the Object/Satellite Catalog (SATCAT), the model uses orbital physics parameters to distinguish between these classes. The `spacetrack` Python library was used as a wrapper for the Space-Track.org REST API, serving as a more intuitive Pythonic interface. The resulting `sklearn.ensemble.RandomForestClassifier` demonstrates high predictive power, reliably achieving an F_1 score above 0.95 for the Debris and Satellite (denoted in the dataset as Payload) categories.

Project Definition

The problem addressed by this code is whether there are differences in the fundamental physics parameters that define the orbits of satellites (payload) and debris. Furthermore, this paper investigates which of these parameters, if any, are the most important and effective at distinguishing between the two classes.

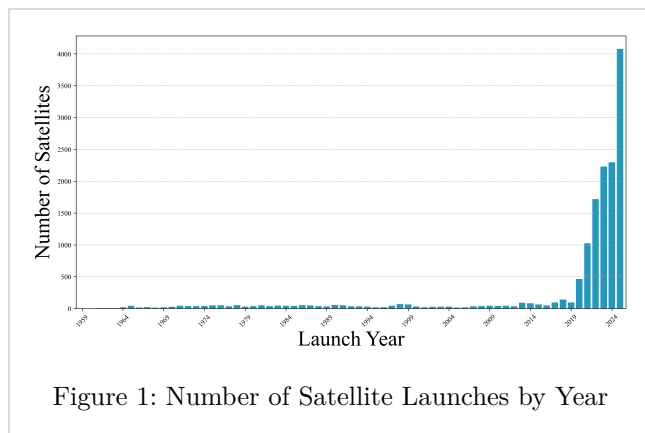


Figure 1: Number of Satellite Launches by Year

Debris represents the second largest class of objects orbiting Earth (Figure 2 (a)). This problem is becoming increasingly important as the number of satellite launches per year grows exponentially, as shown in Figure 1. With this growth, especially in Low Earth Orbits, space is becoming increasingly crowded with satellites and cluttered with debris. Detecting and classifying debris is the first step in addressing space clutter and possibly exploit-

ing the unique orbital characteristics of debris to inform solutions to this problem.

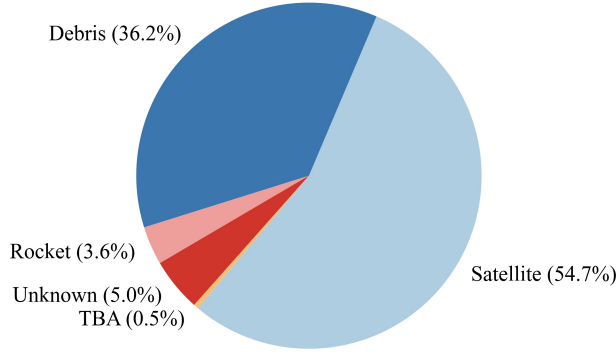
Furthermore, as our night skies become increasingly speckled with the bright dots of satellites, and as satellite construction and launch costs keep decreasing, over time, it will be easier for nefarious actors to deploy satellites for non-peaceful purposes without the general public's knowledge. The author's belief reflects that of generations of policymakers, as documented in the 1967 UN Outer Space Treaty: Space must be used and safeguarded for peaceful purposes and for the benefit of all mankind.

This vision demands international collaboration, and while the US Space Force generously provides information on detected objects in orbit, ensuring the democratic global understanding of the space above all of us should neither be the responsibility nor the purview of any one nation. In the absence of a larger international body, this responsibility falls upon laymen worldwide. It is the right of every citizen of this globe to have access to all the information regarding these objects. This code aims to be the first step in a larger project to visually identify, catalog, and classify satellites in an open-source manner, in an effort to build a global open-source community aimed at democratizing space.

Data Selection and Preprocessing

The data for this study are sourced from the Object/Satellite Catalog (SATCAT) via Space-Track.org. This dataset is maintained by the U.S. Space Force's 18th Space Defense Squadron and is derived from the United States Space Surveillance Network (SSN), a global ar-

(a) Distribution of Objects by Type



(b) Distribution of Satellites by Orbit Type

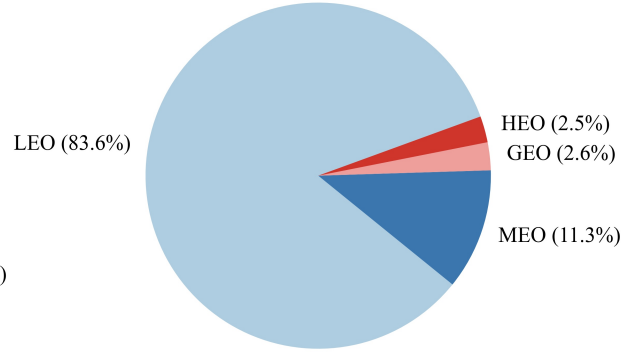


Figure 2: Orbital population analysis: Categorical breakdown. Data sourced from SATCAT via Space-Track.org U.S. Space Force [2024]. Figure generated by author.

ray of sensors to track everything from massive active payloads to fragments as small as a softball. This source was selected for its historical reliability and the ability to perform relatively frequent API queries to maintain an up-to-date catalog.

As shown in Figure 2 (a) above, this dataset contains approximately 50,500 entries across five primary categories:

- **Payload:** Active or inactive satellites.
- **Debris:** Fragments resulting from collisions, degradation, or space operations.
- **Rocket Body:** Expended rocket components, usually upper stages that retained enough energy to remain in orbit.
- **Unknown:** Objects of undetermined origin.
- **TBA:** Newly detected objects awaiting formal cataloging.

To ensure the model learns from physical characteristics rather than administrative metadata, six columns (NORAD_CAT_ID, GP_ID, OBJECT_ID, OBJECT_NAME, and FILE) out of forty were dropped. These columns contain no physical information and could lead to accidental data leakage, causing the model to simply memorize names or IDs rather than investigate the physics of the orbit.

The exponential growth trend displayed in Figure 1 marks a historic change in the availability of cheap and rapid launch options from private providers, primarily SpaceX. Most satellites launched since 2019 have been part of SpaceX’s Starlink constellation, which provides reliable internet access anywhere in the world. Because

Starlink satellites are deployed in Low Earth Orbit (LEO) and currently represent the majority of all satellites ever launched, the vast majority of the tracked orbital population is concentrated in LEO (Figure 2 (b)).

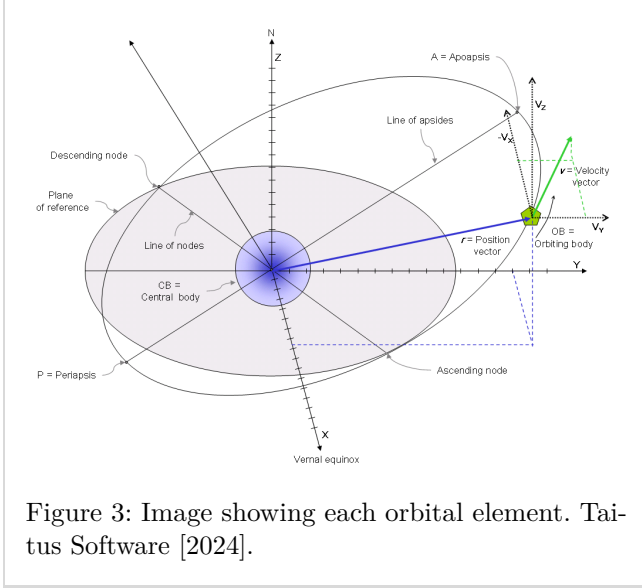
While orbits can be classified by a variety of geometric parameters, the most conventional classification system relies on the orbital period. This is physically grounded in Kepler’s Third Law ($T^2 \propto a^3$), which dictates that an object’s period is strictly determined by its distance from the Earth’s center of mass. This system generally divides orbits into Low Earth Orbit (LEO), Medium Earth Orbit (MEO), and Geostationary Orbit (GEO). See the breakdown of orbital periods in Table 1 below.

Table 1: Classification of Satellite Orbits by Orbital Period

Orbit	Period (min)	Example Use
Low Earth	84 – 128	Starlink
Medium Earth	128 – 1,436	GPS
Geostationary	$\approx 1,436$	Weather

An orbit is uniquely defined in 3D space by six Classical Orbital Elements (COEs): the semi-major axis and eccentricity (defining the size and shape), and the inclination, longitude of the ascending node, and argument of periapsis (defining the orientation of the orbital plane and the ellipse). The final parameter, true anomaly, specifies the object’s instantaneous position. These parameters are visualized in Figure 3. The initial hypothesis was that one or more of these parameters would provide a strong predictive geometric description, and the Random Forest model would thus identify some of these specific elements

as the most effective splitting criteria to maximize the information gain when classifying satellites and debris.



Data Mining Methods

The Random Forest Classification algorithm was selected for this task. A Random Forest is an ensemble learning method that operates by aggregating the predictions of a group of individual decision tree classifiers to achieve greater predictive accuracy. This algorithm typically utilizes a bagging process of training each tree on a unique, random subset of the data. In this manner, the Tree Classifiers search for the optimal split among a random subset of features rather than the entire set, reducing the risk of overfitting. The ensemble then combines these individual outputs via a majority vote, known as a hard voting classifier, which effectively balances bias and variance to yield a more robust and reliable final prediction.

Random Forest is particularly well-suited for this problem because it excels at identifying non-linear relationships between features and provides a clear feature-importance ranking, calculating which physics parameters (such as Eccentricity or Ballistic Coefficient) the model found most revealing.

A significant challenge in this dataset is the class imbalance. As shown in the category breakdown, Debris and Payloads dwarf the Rocket Body, Unknown, and TBA categories. To prevent the model from becoming biased toward the majority classes, the `class_weight='balanced'` hyperparameter was utilized. This technique is most commonly used to handle imbalanced datasets by automatically adjusting weights inversely proportional to class frequencies. The weight for each class using this hyperparameter settings can be

calculated using this formula from Scikit-learn developers [2024]:

$$w_j = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_j} \quad (1)$$

- w_j : Weight assigned to class j .
- n_{samples} : Total number of samples in the dataset.
- n_{classes} : Total number of unique classes.
- n_j : Count of samples belonging to class j .

To investigate the effect of the imbalanced dataset on model predictive power, two Random Forest models were constructed:

1. A comprehensive model including all five categories with weights inversely proportional to the frequency of the given category.
2. A reduced model focusing specifically on the Debris and Payload classes to maximize the precision of the primary classification task.

The implementation used `scikit-learn`, with 500 estimators (decision trees) and the `n_jobs=-1` parameter to utilize all available processor cores.

Evaluation and Results

The performance of the Random Forest models was evaluated using standard classification metrics: precision, recall, and the F_1 -score. By comparing the General Model (which includes all five SATCAT categories) against the Reduced Model (containing solely Debris and Payload data), the effect of statistical noise from minor categories on the models' predictive ability was investigated.

Comparative Performance Analysis

The General Model achieved respectable F_1 scores for identifying Debris (F_1 : 0.93) and Payloads (F_1 : 0.95). However, a closer look at the class-level metrics reveals that while setting the class weight hyperparameter to **balanced** did improve the performance of this model with the smaller classes, it did not fully resolve the issues associated with the imbalanced dataset. The model therefore struggles with classifying the Rocket Body, TBA, and Unknown categories, as demonstrated by the low F_1 scores of 0.44, 0.40, and 0.57 respectively.

Reducing the dataset to just payload and debris improved the model's performance, raising the F_1 score for Debris from 0.93 to 0.97 and for Payload from 0.95 to 0.98 (Table 2). This was primarily achieved by drastically

Table 2: Model Debris/Payload Prediction Comparison

Model Variant	Class	Precision	Recall	F_1 -Score
5-Class	Debris	0.89	0.96	0.93
	Payload	0.93	0.97	0.95
2-Class	Debris	0.97	0.96	0.97
	Payload	0.98	0.98	0.98

increasing the precision for both classes. This confirms that the minor categories in the SATCAT dataset act as statistical noise, mostly increasing the number of misclassifications whilst not significantly affecting the recall score. Therefore, for a deployed model with the primary purpose of differentiating debris from satellites, removing these extra categories is recommended.

Feature Importance and Physical Intuition

To understand how the model distinguishes between these objects, the feature importance rankings were examined using the `rnd_clf.feature_importances_` function. As expected, the results closely align with the aforementioned hypothesis based on the laws of orbital mechanics, with two out of three top Predictor features being among the 6 elements that define a geometric orbit, with the third, the Ballistic Coefficient (B^*), being an important flight drag term.

Top Predictors:

- **Eccentricity (e):** 0.148
- **Ballistic Coefficient (B^*):** 0.128
- **Inclination (i):** 0.120

The prominence of Eccentricity as the top feature is scientifically intuitive. Active satellites (Payloads) typically use onboard propulsion and thrusters to maintain stable, circularized orbits, resulting in an average eccentricity of approximately 0.0012. In contrast, debris fragments are subject to uncontrolled energy loss and gravitational perturbations over time. This leads to more eccentric, elliptical orbits, reflected in the significantly higher average eccentricity of 0.0137.

Table 3: Average Orbital Parameters and Coefficients

Parameter	Debris (Average)	Payload (Average)
Eccentricity	0.014	0.0012
Ballistic Coefficient	0.0018	-1.6×10^{-5}
Inclination	88.5°	63.2°

The data also reveals a stark difference in the Ballistic Coefficient, which measures how an object is affected by atmospheric drag. Payloads showed a negligible average

B^* , whereas Debris showed a much higher coefficient (0.0018). A possible explanation for this phenomenon is that the debris often consists of high-surface-area fragments that are more susceptible to drag than dense, compact satellites designed to minimize drag.

It is important to note that the ballistic coefficient and inclination stochastically switch in feature importance, while eccentricity remains the top feature, once again demonstrating that the orbit’s eccentricity is the best and most stable feature for this classification task.

Strengths and Limitations

The primary strength of this model is its ability to achieve near-perfect classification using few orbital parameters, demonstrating that there is a physical and observable difference between an average piece of debris and an average satellite in a Low Earth Orbit.

However, a notable limitation is the Rocket Body, Unknown, and TBA categories. The model’s poor performance in these areas, due to significantly lower recall, highlights the difficulty of classifying objects that are quantitatively less present in the dataset than the two large Debris and Payload categories. For the open-source application outlined in the Project Definition section, this model provides a highly reliable filter for the vast majority of tracked objects, even if newly detected fragments and small amounts of other objects currently remain ambiguous with these models.

Ethical Considerations

Transparency vs. National Privacy

A primary ethical dilemma lies in the balance between public transparency and the privacy of sovereign states. While this model is designed to distinguish between debris and payloads, the same methodology could, in theory, be used to unmask satellites that a nation might prefer to keep discreet for defensive purposes. However, the author maintains that as space is a global commons, the physical location and general nature of objects in orbit should be public knowledge. Providing laymen with the tools to independently verify the nature of these objects is an act of democratic accountability, ensuring that no single entity can single-handedly decide what remains hidden in our skies.

Potential for Misuse

While this classification model is intended for environmental stewardship and public education, the ability to rapidly catalog active payloads could be exploited for interference or counter-space operations. Currently, most entities with the capability to act nefariously in

space already possess the infrastructure to obtain this information. Thus, new actors do not presently pose a significant threat. However, as launch and manufacturing costs plummet, enabling even university groups to deploy CubeSats, the accessibility of space increases the potential for misuse. The ethical response to this evolving risk is the democratization of information rather than its restriction. By open-sourcing these classification tools, the global community is empowered to collectively monitor orbital activity, utilizing transparency as a vital deterrent against covert and clandestine operations.

Stewardship of the Orbital Environment

Finally, there is an ethical obligation to the environment itself. By accurately identifying debris, we provide the data necessary for future generations to clean the clutter humanity has created. Treating space as a landfill violates the inter-generational trust enshrined in the 1967 Outer Space Treaty. This project serves as a humble investigation aimed at addressing this issue, a small but necessary step in fulfilling our role as responsible stewards of space, ensuring that it remains accessible for the benefit of all mankind.

Future Developments

While the current Random Forest approach provides high accuracy for primary categories, its functionality could be further enhanced in these ways:

- **Integration of TLE Time-Series Data:** The current model uses snapshots of orbital parameters. Future iterations could investigate Two-Line Element (TLE) history to analyze how permutations affect orbits and investigate an object's orbit decays over time, providing more data that could hypothetically be used to better distinguish between satellites and debris.
- **Neural Network Architectures:** While Random Forests are highly interpretable, a Deep Neural Network (DNN) might better capture the high-dimensional relationships between obscure physics parameters, potentially improving the recall for the smaller classes.
- **Real-Time API Integration:** Transitioning the code into a live-tracking dashboard would allow the development of a platform that will send real-time classification alerts as new objects are cataloged by the SSN.
- **Visual Tracking:** The larger goal of this project is to create open-source hardware and software that can detect man-made orbiting objects from Earth

by differentiating them from other celestial objects through their high velocity. A visual model such as Keras could be trained to visually classify the objects as satellites, debris, or something else. The program could also use the Two-Line Element data to associate which satellite is in line of sight based on approximations from the TLE data. The program could also use advanced optical features like voxel projection to detect fainter satellites, such as spy satellites or those in higher orbits.

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